

GROUP - 66 Simple Smart Loader Documentation

Steps to run on terminal:

1. Path of the file
2. Make
3. ./SmartLoader sum
4. Process paging info will be displayed on terminal

Implementation:

- Lazy loading (mapping pages on demand) done by SIGSEGV
- Uninitialized data (BSS) handled using MAP_ANONYMOUS in mmap
- Page faults, allocated pages and internal fragmentation tracked and summary report printed
- On page access (SIGSEGV):
 1. Calculate faulting page_start = fault_addr / PAGE_SIZE * PAGE_SIZE
 2. Identify ELF segment corresponding to fault_addr
 3. Determine protection flags using segment flags
 4. Map page using mmap
 - > From file if inside p_filesz
 - > Anonymous for BSS (p_memsz > p_filesz)
 5. Store page info in pages[]
 6. Update fragmentation
- Proper cleanup performed. And error handling implemented for file validity, file open/read/close, segmentation fault, mmap, sigaction and user input

Github repository link: <https://github.com/antelope0112/SmartLoader-Group66>